

Maven – Final Armageddon

1. Containerizing a Java Web Application Using Maven and Docker

```
Volume 01: Part 19: Maven - 1/19
Directory of D:\maven-docker-2
12-05-2026  22:27    <DIR>          .
12-05-2026  22:26          84 dockerfile
12-05-2026  22:17       1,830 pom.xml
12-05-2026  22:17    <DIR>          src
12-05-2026  22:21    <DIR>          target
                2 File(s)        1,914 bytes
                3 Dir(s)  137,629,466,624 bytes free
D:\maven-docker-2>
```

```
Command Prompt - mvnw.cnr  +  v
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/commons/commons-parent/56/commons-parent-56.pom
(82 kB at 867 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/com/github/luben/zstd-jni/1.5.5-2/zstd-jni-1.5.5-2.pom
Downloaded from central: https://repo.maven.apache.org/maven2/com/github/luben/zstd-jni/1.5.5-2/zstd-jni-1.5.5-2.pom (1.
9 kB at 29 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/commons-io/commons-io/2.13.0/commons-io-2.13.0.jar
Downloaded from central: https://repo.maven.apache.org/maven2/commons-io/commons-io/2.13.0/commons-io-2.13.0.jar (484 kB
at 3.4 MB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-archiver/4.7.1/plexus-archiver
-4.7.1.jar
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/commons/commons-compress/1.23.0/commons-compre
ss-1.23.0.jar
Downloading from central: https://repo.maven.apache.org/maven2/com/github/luben/zstd-jni/1.5.5-2/zstd-jni-1.5.5-2.jar
Downloaded from central: https://repo.maven.apache.org/maven2/com/github/luben/zstd-jni/1.5.5-2/zstd-jni-1.5.5-2.jar (1.
1 MB at 1.0 MB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-archiver/4.7.1/plexus-archiver-
4.7.1.jar (221 kB at 1.1 MB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/commons/commons-compress/1.23.0/commons-compres
s-1.23.0.jar (1.1 MB at 1.0 MB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/com/github/luben/zstd-jni/1.5.5-2/zstd-jni-1.5.5-2.jar (5.
9 MB at 5.4 MB/s)
[INFO] Packaging webapp
[INFO] Assembling webapp [java-web-app] in [D:\maven-docker-2\target\myapp]
[INFO] Processing war project
[INFO] Copying webapp resources [D:\maven-docker-2\src\main\webapp]
[INFO] Building war: D:\maven-docker-2\target\myapp.war
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 15.225 s
[INFO] Finished at: 2026-05-12T22:21:38+05:30
[INFO] -----
```

```
D:\maven-docker-2>type dockerfile
FROM tomcat:10-jdk17
COPY target/myapp.war /usr/local/tomcat/webapps/
EXPOSE 8080
D:\maven-docker-2>
```

```
Command Prompt - mvnw.cnr x + v
D:\maven-docker-2>docker build -t dockermaven .
[+] Building 167.9s (8/8) FINISHED                                docker:desktop-linux
=> [internal] load build definition from dockerfile                0.1s
=> => transferring dockerfile: 121B                                0.1s
=> [internal] load metadata for docker.io/library/tomcat:10-jdk17 5.7s
=> [auth] library/tomcat:pull token for registry-1.docker.io      0.0s
=> [internal] load .dockerignore                                  0.1s
=> => transferring context: 2B                                       0.0s
=> [internal] load build context                                  0.2s
=> => transferring context: 1.70kB                                   0.1s
=> [1/2] FROM docker.io/library/tomcat:10-jdk17@sha256:8ea3c0c53a1388cc53856cda2c523ed081a5331558452f3f90f97bd 160.8s
=> => resolve docker.io/library/tomcat:10-jdk17@sha256:8ea3c0c53a1388cc53856cda2c523ed081a5331558452f3f90f97bd47 0.0s
=> => sha256:4c638352d0183234b4c73b07f702e5f49e508c4f35881b94f2d0dd2eaf604f79 15.00MB / 15.00MB 54.4s
=> => sha256:9a82482d42313167ff20e2183b6c787c9d6dc9e48b6613e71fa4eb6937ac5ea3 22.97MB / 22.97MB 136.9s
=> => sha256:fb57b967c0a732556d1c785c001da6e4ef2644fa605d970685bf28ef77ed2a84 2.28kB / 2.28kB 2.1s
=> => sha256:ef8b71cc80fb97353cbdebfd03f5ad4f853c3c714101adbf2f9b6cd6bb2571bd 158B / 158B 1.8s
=> => sha256:3d963415bc0793892dc201ad765ab380ae0fa4dcd110b7704b502ca3a13ddbc4 145.91MB / 145.91MB 155.8s
=> => sha256:a3bb4f9eb54385d362f265545d0d33ca19f6e42165c6e2e8c39864b057bbdf48 139B / 139B 1.5s
=> => extracting sha256:9a82482d42313167ff20e2183b6c787c9d6dc9e48b6613e71fa4eb6937ac5ea3 1.4s
=> => extracting sha256:3d963415bc0793892dc201ad765ab380ae0fa4dcd110b7704b502ca3a13ddbc4 2.4s
=> => extracting sha256:ef8b71cc80fb97353cbdebfd03f5ad4f853c3c714101adbf2f9b6cd6bb2571bd 0.0s
=> => extracting sha256:fb57b967c0a732556d1c785c001da6e4ef2644fa605d970685bf28ef77ed2a84 0.0s
=> => extracting sha256:a3bb4f9eb54385d362f265545d0d33ca19f6e42165c6e2e8c39864b057bbdf48 0.0s
=> => extracting sha256:4f4fb700ef54461cfa02571ae0db9a0dc1e0cdb557748a6d75e68dc38e8acc1 0.0s
=> => extracting sha256:4c638352d0183234b4c73b07f702e5f49e508c4f35881b94f2d0dd2eaf604f79 0.3s
=> [2/2] COPY target/myapp.war /usr/local/tomcat/webapps/        0.4s
=> => exporting to image                                             0.4s
=> => exporting layers                                              0.1s
=> => exporting manifest sha256:33886e5f27427dcfb29970a56331ac30d429415e22628b2376be9e1f9be17dd9 0.0s
=> => exporting config sha256:05ea68a0d323e7d66881f4ae083d8b0601b233271c910a4215e9c634f9938faa 0.0s
=> => exporting attestation manifest sha256:6109636ef6fe8e648fd94acd2583a3919f68221162cc4b2b43b5cf48ec363a3f 0.1s
=> => exporting manifest list sha256:899880dc8803525cf11ff9cd7056adffd18a60d08373ce47ac54e8532a89922 0.0s
=> => naming to docker.io/library/dockermaven:latest              0.0s
=> => unpacking to docker.io/library/dockermaven:latest           0.0s

D:\maven-docker-2>docker run -dit --name testmaven -p 8083:8080 dockermaven
42e7a584e8403398a7bfa1f54d0e010612f674f264e55df7bd44a6dced6f21f0
```

```
=> [2/2] COPY target/myapp.war /usr/local/tomcat/webapps/        0.4s
=> => exporting to image                                             0.4s
=> => exporting layers                                              0.1s
=> => exporting manifest sha256:33886e5f27427dcfb29970a56331ac30d429415e22628b2376be9e1f9be17dd9 0.0s
=> => exporting config sha256:05ea68a0d323e7d66881f4ae083d8b0601b233271c910a4215e9c634f9938faa 0.0s
=> => exporting attestation manifest sha256:6109636ef6fe8e648fd94acd2583a3919f68221162cc4b2b43b5cf48ec363a3f 0.1s
=> => exporting manifest list sha256:899880dc8803525cf11ff9cd7056adffd18a60d08373ce47ac54e8532a89922 0.0s
=> => naming to docker.io/library/dockermaven:latest              0.0s
=> => unpacking to docker.io/library/dockermaven:latest           0.0s

D:\maven-docker-2>docker run -dit --name testmaven -p 8083:8080 dockermaven
42e7a584e8403398a7bfa1f54d0e010612f674f264e55df7bd44a6dced6f21f0
```



Hello from Docker + Maven!

2. Containerizing and Deploying a Java Web Application Using Maven Archetype, WAR Packaging, Jetty, and Docker

```

Command Prompt - mvnw.cnr  X + v
D:\jetty-maven>mvn archetype:generate -DgroupId=com.example -DartifactId=mywebapp -DarchetypeArtifactId=maven-archetype-
webapp -DarchetypeVersion=1.4 -DinteractiveMode=false
[INFO] Scanning for projects...
Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/mojo/maven-metadata.xml
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-metadata.xml
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-metadata.xml (14 kB at 16 k
B/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/mojo/maven-metadata.xml (21 kB at 22 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-archetype-plugin/maven-met
adata.xml
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-archetype-plugin/maven-met
adata.xml (1.1 kB at 5.8 kB/s)
[INFO]
[INFO] -----< org.apache.maven:standalone-pom >-----
[INFO] Building Maven Stub Project (No POM) 1
[INFO] -----[ pom ]-----
[INFO]
[INFO] >>> archetype:3.4.1:generate (default-cli) > generate-sources @ standalone-pom >>>
[INFO]
[INFO] <<< archetype:3.4.1:generate (default-cli) < generate-sources @ standalone-pom <<<
[INFO]
[INFO]
[INFO] --- archetype:3.4.1:generate (default-cli) @ standalone-pom ---
[INFO] Generating project in Batch mode
Downloading from central: https://repo.maven.apache.org/maven2/archetype-catalog.xml
Downloaded from central: https://repo.maven.apache.org/maven2/archetype-catalog.xml (18 MB at 8.7 MB/s)
[INFO]
[INFO] -----
[INFO] Using following parameters for creating project from Archetype: maven-archetype-webapp:1.4
[INFO] -----
[INFO] Parameter: groupId, Value: com.example

```

```

Command Prompt - mvnw.cnr  X + v
[INFO] -----
[INFO] Using following parameters for creating project from Archetype: maven-archetype-webapp:1.4
[INFO] -----
[INFO] Parameter: groupId, Value: com.example
[INFO] Parameter: artifactId, Value: mywebapp
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] Parameter: package, Value: com.example
[INFO] Parameter: packageInPathFormat, Value: com/example
[INFO] Parameter: package, Value: com.example
[INFO] Parameter: groupId, Value: com.example
[INFO] Parameter: artifactId, Value: mywebapp
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] Project created from Archetype in dir: D:\jetty-maven\mywebapp
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 12.926 s
[INFO] Finished at: 2026-05-12T23:00:24+05:30
[INFO] -----

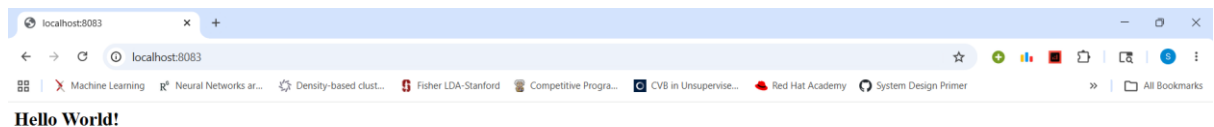
D:\jetty-maven>cd mywebapp

D:\jetty-maven\mywebapp>mvn clean package
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.example:mywebapp >-----
[INFO] Building mywebapp Maven Webapp 1.0-SNAPSHOT
[INFO] from pom.xml
[INFO] -----[ war ]-----
[INFO]

```

```
Command Prompt - mvnw.cr x + v
D:\jetty-maven\mywebapp>docker build -t mavenjetty .
[+] Building 41.9s (8/8) FINISHED
=> [internal] load build definition from dockerfile
=> => transferring dockerfile: 127B
=> [internal] load metadata for docker.io/library/jetty:11-jdk17
=> [auth] library/jetty:pull token for registry-1.docker.io
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [internal] load build context
=> => transferring context: 2.14kB
=> [1/2] FROM docker.io/library/jetty:11-jdk17@sha256:2da9052881dfcfa9b08cd2b025a411c96a4466f40b9c7d2ded720df4dd
=> => resolve docker.io/library/jetty:11-jdk17@sha256:2da9052881dfcfa9b08cd2b025a411c96a4466f40b9c7d2ded720df4dd
=> => sha256:682d1a3197c6e11de7df99dd2f999e6acfbca27646f512103d6a81c3d3994cf 159B / 159B
=> => sha256:cda4cc421c2702efa035a5cd9d3abe9ece7b0ef008b8ab57e85d58ba2685eab1 1.84kB / 1.84kB
=> => sha256:3404c492510be2396378b473c5470243eb685ac3c2fdcef9fbb84ee599ddc259 144.85MB / 144.85MB
=> => sha256:c940cc503ed4660889aab5fe6ab7b4e7f6f5668ee17a74fb1b5918d33e645f92 2.28kB / 2.28kB
=> => sha256:5dc028e251945806299735ec40f07449468572c1ccfbd2996ed8a905d06a2e86 14.70MB / 14.70MB
=> => sha256:f72f0f1ec9ea380a6af4a85e594cfff7bccbe1c7d5311177ebdc52d3e6d2f04b7 22.96MB / 22.96MB
=> => extracting sha256:f72f0f1ec9ea380a6af4a85e594cfff7bccbe1c7d5311177ebdc52d3e6d2f04b7
=> => extracting sha256:3404c492510be2396378b473c5470243eb685ac3c2fdcef9fbb84ee599ddc259
=> => extracting sha256:682d1a3197c6e11de7df99dd2f999e6acfbca27646f512103d6a81c3d3994cf
=> => extracting sha256:c940cc503ed4660889aab5fe6ab7b4e7f6f5668ee17a74fb1b5918d33e645f92
=> => extracting sha256:5dc028e251945806299735ec40f07449468572c1ccfbd2996ed8a905d06a2e86
=> => extracting sha256:4f4fb700ef54461cfa02571ae0db9a0dc1e0cbb5577484a6d75e68dc38e8acc1
=> => extracting sha256:cda4cc421c2702efa035a5cd9d3abe9ece7b0ef008b8ab57e85d58ba2685eab1
=> [2/2] COPY target/mywebapp.war /var/lib/jetty/webapps/root.war
=> => exporting to image
=> => exporting layers
=> => exporting manifest sha256:6a1e858e003b6187668c8ad3bee50b79c1277fda9a7de9f98b969697c81a5db8
=> => exporting config sha256:f0403dc7d265e4808f893bcabe98f2ef4efda5ef4d19b229f3205fe64455fba4
=> => exporting manifest sha256:6a1e858e003b6187668c8ad3bee50b79c1277fda9a7de9f98b969697c81a5db8
=> => exporting config sha256:f0403dc7d265e4808f893bcabe98f2ef4efda5ef4d19b229f3205fe64455fba4
=> => exporting attestation manifest sha256:5c9892b79a05985cd6e95508653bdbce0a729e66eed7f36c193a0e086deed2a3
=> => exporting manifest list sha256:c88b61227d1e35ee2ee3ec058754ab2fec6a0fefe7c6e93280361c4c33014ec3
=> => naming to docker.io/library/mavenjetty:latest
=> => unpacking to docker.io/library/mavenjetty:latest

D:\jetty-maven\mywebapp>docker run -d -p 8083:8080 --name jetty-mav mavenjetty
c142fe3452a0a343d3ee0d18eb447018f5eeba26e5dc2803813977b507afad0
```



More production-grade dockerfile:

```
Command Prompt - mvnw.cr x + v
D:\jetty-maven\mywebapp>dir
Volume in drive D is New Volume
Volume Serial Number is AEE6-4A19

Directory of D:\jetty-maven\mywebapp

12-05-2026 23:03 <DIR>      .
12-05-2026 23:00 <DIR>      ..
12-05-2026 23:21             241 dockerfile
12-05-2026 23:00             2,265 pom.xml
12-05-2026 23:00 <DIR>      src
12-05-2026 23:01 <DIR>      target
                2 File(s)      2,506 bytes
                4 Dir(s)  137,629,458,432 bytes free

D:\jetty-maven\mywebapp>type dockerfile
FROM maven:3.9.9-eclipse-temurin-17 AS builder
WORKDIR /app
COPY pom.xml .
COPY src ./src
RUN mvn clean package -DskipTests
FROM jetty:11-jdk17
COPY --from=builder /app/target/mywebapp.war /var/lib/jetty/webapps/root.war
EXPOSE 8080

D:\jetty-maven\mywebapp>
```